HS017

HS Risk management form

For additional information refer to HS329 Risk Management Procedure

Faculty/Division: Faculty of Engineering		School/Unit: School of Electrical Engineering and Telecommunications		
Document number	Initial Issue date 18 November, 2025	Current version	Current Version Issue date	Next review date

Risk management name Risk Assessment for ISAC Related Simulation and Research

Form completed by	Xinpeng Yu	*Signature*	*Date* 18/11/2025
Responsible supervisor/ authorising officer		*Signature*	*Date*

Identify the activity and the location of the activity

Description of activity

Developing Cell-Free mMIMO ISAC related algorithms.
Simulations based on MATLAB,
Writing results report and poster.

Description of location

UNSW Kensington campus (EET Building, Main Library, Law Library, Home

Identify who may be at risk from the activity:

This may include fellow workers, visitors, contractors and the public. The types of people may affect the risk controls needed and the location may affect the number of people at risk

Persons at risk

XInpeng Yu (student), Derrick Ng (supervisor), Shaokang Hu (co-supervisor)

How they were consulted on the risk

Safety briefing via course guidelines & Supervisor discussion

List legislation, standards, codes of practice, manufacturer's guidance etc used to determine control measures necessary
Work Health and Safety Act 2011 Work Health and Safety Regulation 2011

Identify hazards and control the risks.
1. An activity may be divided into tasks. For each task identify the hazards and associated risks. Also list the possible scenarios which could sooner or later cause harm.
2. Determine controls necessary based on legislation, codes of practice, Australian standards, manufacturer's instructions etc.
3. List existing risk controls and any additional controls that need to be implemented
4. Rate the risk once all controls are in place using the matrix in HS329 Risk Management Procedure

SHADED GREY AREAS
If you need to determine whether it's reasonably practicable to implement a control, based on the risk complete the shaded grey columns

Feel free to resize the boxes to suit your situation/the amount of text you need to use

Task/ Scenario	*Hazard*	*Associated harm*	*Existing controls*	*Any additional controls required?*	*Risk Rating*			*Cost of controls (in terms of time, effort, money)*	*Is this reasonably practicable* Y/N
					C	*L*	*R*		
Coding and algorithm development	Long time using computer, Long time seated	Eye strain, Lumbar muscle strain	• Regular breaks Limited daily work time		2	1	2	RP	Y
MATLAB Simulation	Misuse of pirated software	Harm to university's reputation	• Only use the genuine software provided by the university		3	1	4	N/A	Y
			•						
			•						

RISK RATING METHODOLOGY AND MATRIX

Consider the Consequences

Consider: What type of harm could occur (minor, serious, death)? Is there anything that will influence the severity (e.g. proximity to hazard, person involved in task etc.). How many people are exposed to the hazard? Could one failure lead to other failures? Could a small event escalate?

5. Severe: death or permanent disability to one or more persons

4. Major: hospital admission required

3. Moderate: medical treatment required

2. Minor: first aid required

1. Insignificant: injuries not requiring first aid

Consider the Likelihood

Consider: How often is the task done? Has an accident happened before (here or at another workplace)? How long are people exposed? How effective are the control measures? Does the environment effect it (e.g. lighting/temperature/pace)? What are people's behaviours (e.g. stress, panic, deadlines) What people are exposed (e.g. disabled, young workers etc.)?

A. Almost certain: expected to occur in most circumstances

B. Likely: will probably occur in most circumstances

C. Possible: might occur occasionally

D. Unlikely: could happen at some time

E. Rare: may happen only in exceptional circumstances

Calculate the Risk

1.Take the consequences rating and select the correct column

2.Take the likelihood rating and select the correct row

3. Select the risk rating where the two ratings cross on the matrix below.

VH = Very high, H = High, M = Medium, L = Low

		CONSEQUENCES				
		1	2	3	4	5
LIKELIHOOD	A	M	H	H	VH	VH
	B	M	M	H	H	VH
	C	L	M	H	H	VH
	D	L	L	M	M	H
	E	L	L	M	M	M

Risk level	Required action
Very high	**Act immediately**: The proposed task or process activity must not proceed. Steps must be taken to lower the risk level to as low as reasonably practicable using the hierarchy of risk controls
High	**Act today:** The proposed activity can only proceed, provided that: (i) the risk level has been reduced to as low as reasonably practicable using the hierarchy of risk controls and (ii) the risk controls must include those identified in legislation, Australian Standards, Codes of Practice etc. and (iii) the document has been reviewed and approved by the Supervisor and (iv) a Safe Working Procedure or Safe Work Method has been prepared and (v) the supervisor must review and document the effectiveness of the implemented risk controls
Medium	**Act this week**: The proposed task or process can proceed, provided that: (i) the risk level has been reduced to as low as reasonably practicable using the hierarchy of controls and (ii) the document has been reviewed and approved by the Supervisor and (iii) a Safe Working Procedure or Safe Work Method has been prepared.
Low	**Act this month:** Managed by local documented routine procedures which must include application of the hierarchy of controls.

List emergency procedures and controls
List emergency controls for how to deal with fires, spills or exposure to hazardous substances and/or emergency shutdown procedures

Implementation			
Additional control measures needed:	**Resources required**	**Responsible person**	**Date of implementation**

REVIEW			
Scheduled review date:			
Are all control measures in place?			
Are controls eliminating or minimising the risk?			
Are there any new problems with the risk?			
Review by: (name)			
Review date:			

Acknowledgement of Understanding

All persons performing these tasks must sign that they have read and understood the risk management (as described in HS329 Risk Management Procedure).

Note: for activities which are low risk or include a large group of people (e.g. open days, BBQ's, student classes etc), only the persons undertaking the key activities need to sign below. For all others involved in such activities, the information can be covered by other methods including for example a safety briefing, induction, and/or safety information sheet (ensure the method of communicating this information is specified here)

Risk management name and version number:	**I have read and understand this risk management form**	
Name	Signature	Date
Xinpeng Yu		18/11/2025